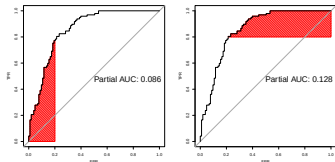
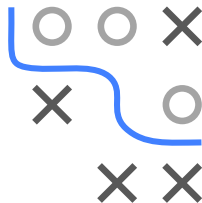


Introduction to Machine Learning

Evaluation Partial AUC

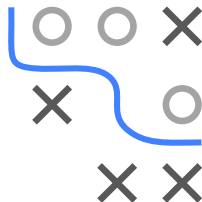


Learning goals

- Understand that entire AUC is not always relevant
- Learn about partial AUC

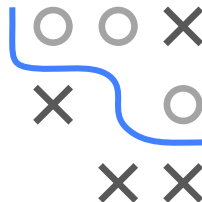
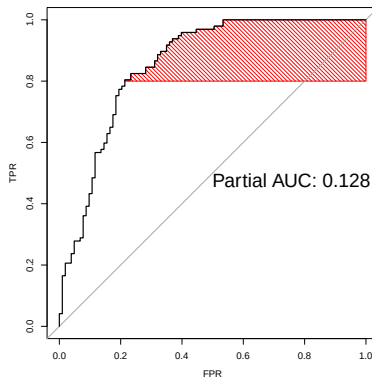
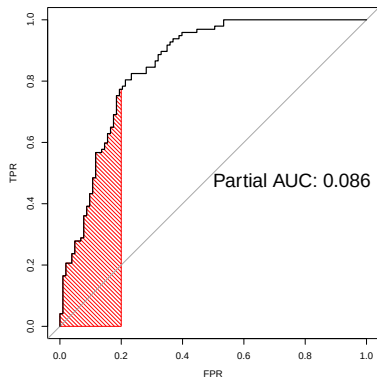
PARTIAL AUC I

- TPR and FPR often treated asymmetrically in biomed contexts
- TPR = disease detection, is crucial
- But low FPR needed to avoid unnecessary treatments
- Common solution: Fix either TPR or FPR to a required value and optimize the other, but not easy to select exact point



PARTIAL AUC II

- Can be useful to limit region under ROC curve
- E.g. $\text{FPR} > 0.2$ or $\text{TPR} < 0.8$ might not be acceptable for task, then we don't want to integrate over that region

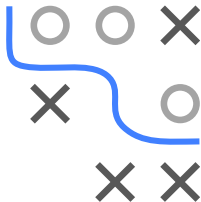
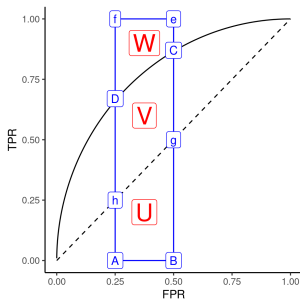


CORRECTED PARTIAL AUC I

- Range of pAUC depends on cut-off values
- Normalize to $[0, 1]$:

$$\text{pAUC}_{\text{corrected}} = \frac{1}{2} \left(1 + \frac{\text{pAUC} - \text{pAUC}_{\min}}{\text{pAUC}_{\max} - \text{pAUC}_{\min}} \right),$$

- $pAUC$ is $V+U = "A-B-C-D"$
- $pAUC_{min}$ is $pAUC$ of random classifier, so $U = "A-B-g-h"$
- $pAUC_{max}$ is $U+V+W = "A-B-e-f"$
- Compute percentage of V in $V+W$
- Rescale so random=0.5; optimal=1



2WAY PARTIAL AUC I

- Can also limit both TPR and FPR
- 2way pAUC = compute area under 2way limited segment

